1 Solution — GIAM 3.3 Problem 2

1.1 Problem

Prove that whenever a prime p does not divide the square of an integer, it also doesn't divide the original integer:

$$p \nmid x^2 \implies p \nmid x.$$

1.2 Answer

$$\boxed{\text{Proved by contraposition: instead show } p \mid x \implies p \mid x^2.}$$

1.3 Proof

Theorem. *For every prime p and every integer x, if $p \nmid x^2$ then $p \nmid x$.*

Proof. *The statement is logically equivalent to its contrapositive,*

$$p \mid x \implies p \mid x^2,$$

so we prove that instead. (Both sides of the original are negations, and $\neg\neg P$ is just P, so the contrapositive is the same statement with the two "not divides" turned into "divides" and swapped.)

Let p be a prime and x an integer, both particular but arbitrarily chosen, and suppose $p \mid x$. By the definition of divisibility there is an integer k with $x = pk$. Squaring,

$$x^2 = (pk)^2 = p^2k^2 = p\left(pk^2\right).$$

> *Set $j = pk^2$. Since p and k are integers and $\mathbb{Z}$ is closed under multiplication, j is an integer, and $x^2 = pj$. By the definition of divisibility, $p \mid x^2$.*
>
> *Since p and x were arbitrary, $p \mid x \implies p \mid x^2$ holds universally, and therefore so does the original statement.* ■

1.4 The Shape of the Argument

TO PROVE **for every prime p and every integer x : $p \nmid x^2 \Rightarrow p \nmid x$**

both sides are negative, so contraposition is the obvious first move

↓

THE CONTRAPOSITIVE — and the whole proof

$p \mid x \Rightarrow p \mid x^2$

Suppose $p \mid x$. Then $x = pk$ for some integer k, so

$$x^2 = (pk)^2 = p^2k^2 = p\cdot(pk^2)$$

and pk^2 is an integer, so $p \mid x^2$. ■

NOTICE WHAT THE PROOF NEVER TOUCHED

Nowhere did the argument use that p is prime. Only "p is an integer" was needed, so the same three lines prove

for every integer $n \geq 2$ and every integer x : $n \nmid x^2 \Rightarrow n \nmid x$

confirmed for every modulus n from 2 to 200 and every x from −300 to 300

SO WHERE IS PRIMALITY ACTUALLY NEEDED? — in the converse

$n \mid x^2 \Rightarrow n \mid x$

This is Euclid's lemma when n is prime, and it is a genuinely deeper fact. For a general modulus it simply fails:

n	x	n divides x^2 ?	n divides x ?
4	2	yes, $x^2 = 4$	no
8	4	yes, $x^2 = 16$	no
9	3	yes, $x^2 = 9$	no
12	6	yes, $x^2 = 36$	no

the converse holds exactly for the squarefree n — checked for every n up to 200; the first failures are 4, 8, 9, 12, 16, 18

One line that settles both directions at once:
writing v(x) for the number of factors of p in x, we have $v(x^2) = 2v(x)$, so $v(x^2) \geq 1$ exactly when $v(x) \geq 1$.

Figure 1.1: A diagram in four bands. The top band states what is to be proved: for every prime p and every integer x, p does not divide x squared implies p does not divide x, noting that both sides are negative so contraposition is the obvious first move. The second band, in green, gives the contrapositive and the whole proof: p divides x implies p divides x squared; suppose p divides x, then x equals pk for some integer k, so x squared equals p squared k squared equals p times pk squared, and pk squared is an integer, so p divides x squared. The third band, in amber, is headed NOTICE WHAT THE PROOF NEVER TOUCHED: nowhere did the argument use that p is prime, only that p is an integer, so the same three lines prove that for every

integer n at least 2 and every integer x, n does not divide x squared implies n does not divide x — confirmed for every modulus from 2 to 200 and every x from minus 300 to 300. The fourth band, in red, asks where primality is actually needed and answers: in the converse, n divides x squared implies n divides x, which is Euclid's lemma when n is prime and a genuinely deeper fact. A table of counterexamples for general moduli lists n equals 4 with x equals 2, n equals 8 with x equals 4, n equals 9 with x equals 3, and n equals 12 with x equals 6; in each row n divides x squared but n does not divide x. The converse holds exactly for the squarefree n, checked for every n up to 200, the first failures being 4, 8, 9, 12, 16, 18. A closing strip notes that writing v of x for the number of factors of p in x, we have v of x squared equals twice v of x, so v of x squared is at least 1 exactly when v of x is at least 1.

1.5 Remark — Why Contraposition Is Forced Here

Some problems merely *benefit* from contraposition. This one all but demands it, and the reason is visible in the shape of the statement: **both** the hypothesis and the conclusion are negative.

A hypothesis of the form $p \nmid x^2$ gives you almost nothing to work with. Divisibility is an *existential* claim — $p \mid x^2$ means "there exists j with $x^2 = pj$" — so its negation says no such j exists. You cannot unpack a non-existence into an equation and start doing algebra with it.

Contraposition converts both negatives into positives at a stroke:

$$\underbrace{p \nmid x^2 \implies p \nmid x}_{\text{no equations available}} \qquad \text{becomes} \qquad \underbrace{p \mid x \implies p \mid x^2}_{\text{hypothesis is an equation}} .$$

Now the hypothesis hands over $x = pk$, which can be squared. This is the same mechanic as in Problem 1, but sharper: there the direct route was merely blocked by an inconvenient cube root, whereas here the direct route has no starting material at all.

The general rule this illustrates: when a conditional has a negated hypothesis, check the contrapositive first. Definitions unpack into equations; the negations of definitions unpack into nothing.

1.6 Remark — The Primality Hypothesis Is Never Used

Read the proof back and look for the word *prime*. It does not appear. The only property of p the argument uses is that it is an integer, so that pk^2 is again an integer. The theorem actually proved is

$$\forall n \in \mathbb{Z},\ \forall x \in \mathbb{Z}, \qquad n \nmid x^2 \implies n \nmid x,$$

verified for every modulus $2 \le n \le 200$ against every x with $|x| \le 300$.

This is the same phenomenon as in 3.1 Problem 8, where the hypothesis $6 \mid (a+b)$ turned out to use nothing but $2 \mid 6$. It is always worth doing this audit at the end of a proof: an unused hypothesis is either a free generalisation or a sign that you have proved the wrong thing.

Here it is the former — and it also tells you something about the *author's* intent. Why state it for primes at all? Because of where the lemma gets used.

1.7 Remark — Where Primality Really Lives: the Converse

The converse of the statement is

$$n \mid x^2 \implies n \mid x,$$

and *this* is where primality is indispensable. It is **Euclid's lemma** in disguise: if p is prime and $p \mid ab$ then $p \mid a$ or $p \mid b$; taking $a = b = x$ gives exactly $p \mid x^2 \Rightarrow p \mid x$.

For a general modulus it is false, and the counterexamples are easy to name:

n	x	$n \mid x^2$?	$n \mid x$?
4	2	yes, $x^2 = 4$	no
9	3	yes, $x^2 = 9$	no
12	6	yes, $x^2 = 36$	no

Exhaustive search over $2 \le n \le 200$ shows the converse holds for a modulus n **exactly when n is squarefree** — the first failures being $4, 8, 9, 12, 16, 18$, each divisible by a square. Primes are squarefree, so the converse holds for them; but so is 6, and the converse holds there too. Primality is sufficient, not necessary — squarefreeness is the exact condition.

So the two directions sit at very different depths. The direction this problem asks for is three lines of definition-chasing valid for any modulus. The reverse direction needs the fundamental theorem of arithmetic (or Bézout) and is genuinely a theorem about primes.

1.8 Remark — Both Directions in One Line, via Valuations

There is a slicker way to see the whole picture. For a prime p, let $v_p(x)$ be the exponent of p in the prime factorisation of x. Then v_p turns multiplication into addition,

$$v_p(ab) = v_p(a) + v_p(b), \qquad \text{so} \qquad v_p\left(x^2\right) = 2\,v_p(x),$$

verified for $p \in \{2,3,5,7,11\}$ over $1 \le x < 400$. Since $p \mid y$ is precisely $v_p(y) \ge 1$,

$$p \mid x^2 \iff 2\,v_p(x) \ge 1 \iff v_p(x) \ge 1 \iff p \mid x,$$

using that $v_p(x)$ is a non-negative *integer*, so $2v_p(x) \ge 1$ forces $v_p(x) \ge 1$. Both directions fall out together, and the exponent 2 is plainly irrelevant: the same line gives $p \mid x^n \iff p \mid x$ for every $n \ge 1$ (checked for $n \le 8$ and all primes below 50).

The catch is that this argument *presupposes* unique factorisation, which is a much heavier tool than the problem intends. The contrapositive proof above is elementary and self-contained; this one is a bird's-eye view of why the fact is true.

1.9 Remark — What This Lemma Is For

Statements like this one look like busywork until you see them deployed. This is precisely the lemma needed to prove that $\sqrt{p}$ is irrational for any prime p:

> *Suppose $\sqrt{p} = a/b$ in lowest terms. Then $a^2 = p\,b^2$, so $p \mid a^2$, hence $p \mid a$ by the converse direction. Writing $a = pc$ gives $p^2c^2 = p\,b^2$, so $b^2 = p\,c^2$, so $p \mid b^2$ and therefore $p \mid b$. But then p divides both a and b, contradicting lowest terms.*

That argument is exactly Theorem 3.3.2 of the text with 2 replaced by an arbitrary prime — and the reason it *needs* a prime, rather than any old n, is the failure documented above. Try to run it with $n = 4$ and it collapses at the first step, which is fortunate, since $\sqrt{4} = 2$ is perfectly rational.